Summary of Topics

1. Scheduling
    - Know how to construct a digraph from a dependency list
    - Know the meaning of the following
        (a) Project digraph
        (b) schedule
        (c) priority list
        (d) processor
        (e) critical path
        (f) critical time
        (g) idle time
        (h) decreasing time algorithm/priority list
        (i) backflow algorithm
        (j) critical time priority list
    - Be able to create a schedule from a digraph or table of dependencies and a priority list, and determine the finishing time, idle time, and whether the schedule is optimal
2. Cryptography
    - Know how to encrypt and decrypt a message using
        - a Caesar cipher
        - an alpha-numeric cipher given a substitution mapping
        - a transposition cipher given the length of a row
        - a transposition cipher given an encryption keyword
        - a shifting substitution / progressive Caesar cipher
        - a Vigenère cipher
        - a double transposition cipher (given two keywords)
    - Know how to use letter frequency to decrypt a message without an encryption key.
    - Know which encryption methods do or do not preserve letter frequency.
    - Be able to articulate strengths and weakness in each of the encryption methods learned.

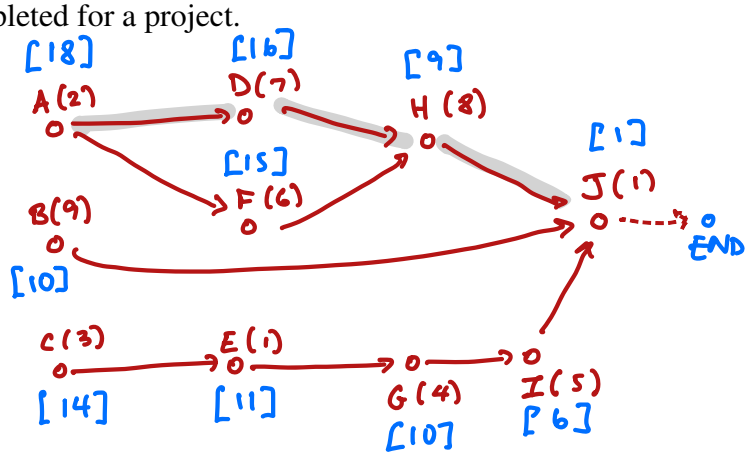
Tabula Recta

A Tabula Recta is provided on the last page. It will be provided on the exam as well.

Scheduling Sample Problems

1. The table below contains the tasks to be completed for a project.

task	time (in minutes)	dependencies/ prerequisites
A	2	
B	9	
C	3	
D	7	A
E	1	C
F	6	A
G	4	E
H	8	D,F
I	5	G
J	1	B,H,I



(a) Next to the table, create a **digraph** representing the project.

(b) Find a **critical path** and its corresponding **critical time**. Explain the significance of the critical time.

Critical path: A → D → H → J

Critical time = 18

Any schedule must take at least 18 time units because the entire critical path must be completed in order.

(c) Create a priority list using the **decreasing time** algorithm. (That is, create a priority list ordered by decreasing time.)

List tasks from longest task time to shortest:

B	H	D	F	I	G	C	A	E	J
9	8	7	6	5	4	3	2	1	1

(d) Use the **Backflow algorithm** to assign critical numbers to each vertex in the graph above.

(e) Create a priority list using the critical time algorithm. (That is, create a priority list ordered by decreasing **critical** time.)

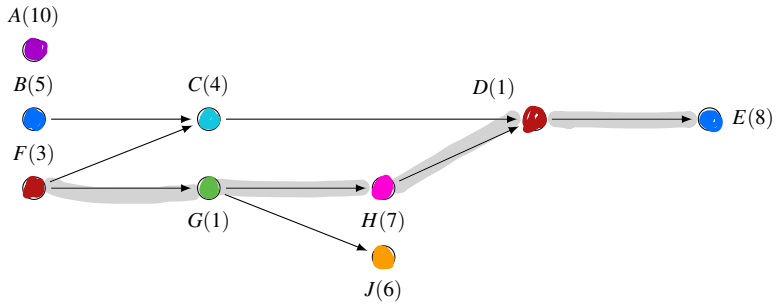
A	D	F	C	E	B	G	H	I	J
18	16	15	14	11	10	10	9	6	1

(f) How long would it take for one processor to complete all the tasks? 46 units

(g) Is it possible for 2 processors to complete a schedule in only the critical time? Why or why not? Provide a supporting calculation.

No, the critical time is 18 but if we divide the total work between the 2 processors evenly we get $\frac{46}{2} = 23$. So someone needs to work at least 23 units.

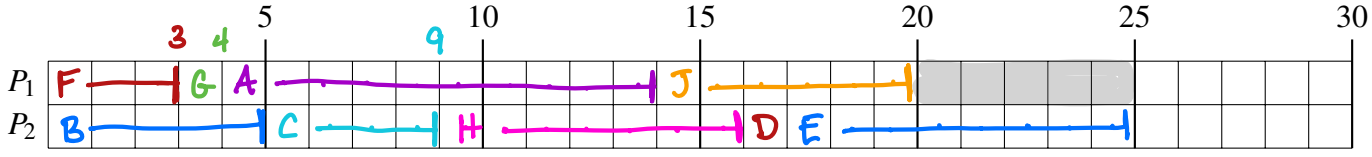
2. Use the diagraph below and the given priority list to construct a schedule using two processors.



Construct a schedule using the priority list

F, B, G, C, A, H, J, D, E

time	0	3	4	5	9	14	16	17	20	25
ready	~~F~~ ~~B~~ A	~~G~~ A	~~A~~ H J	C H J	~~H~~ J D	~~J~~ D	~~D~~	E	—	—
done		F	G	B	C	A	H	D	J	E



Finishing time: 25

Idle time: 5 units

Critical path / time: F → G → H → D → E, 20 time units

Is your schedule optimal? How do you know?

No, the critical time was 20 units but this took 25 units.

(Just for fun, note that the total work time for all the tasks is 45 work units, so if it were possible to split up the work evenly between 2 processors, the best you could do is $\frac{45}{2} = 22.5$ Since we don't allow processors to work half-time units, the best possible finishing time for 2 processors is 23 time units.

The priority list F B G H C J D A E finishes in 23 units; critical path list gives 25, like the list above, and decreasing time takes 27 units.)

Cryptography Sample Problems

1. Encrypt the message GO TO BERLIN using

   (a) a Caesar cipher with a shift of 24 (A → Y)

		A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
0	A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
24	Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X

EMRMZ CPJGL

   (b) a transposition cipher given a row length of 5.

Row length 5 and 10 characters means 2 rows

GOTOB
ERLIN → GEORT LOIBN

   (c) a transposition cipher given encryption keyword STOP.

S	T	O	P
G	O	T	O
B	E	R	L
I	N	P	Q

→

O	P	S	T
T	O	G	O
R	L	B	E
P	Q	I	N

→ TRPOL QGBIO EN

   (d) a shifting substitution cipher (progressive Caesar cipher) starting with a shift of 6 (A to G).

		0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
		A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
6	G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
7	H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
8	I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
9	J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
10	K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
11	L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
12	M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
13	N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
14	O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
15	P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
16	Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P

GOTOB ERLIN
MVBXL PDYWC

MVBXL PDYWC

(e) a Vigenére cipher with key word STOP.

GOTOBERLIN
STOPSTOPST
YHHDTXFAAG

		0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
		A	**B**	**C**	**D**	**E**	**F**	**G**	**H**	**I**	**J**	**K**	**L**	**M**	**N**	**O**	**P**	**Q**	**R**	**S**	**T**	**U**	**V**	**W**	**X**	**Y**	**Z**
18	**S**	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
19	**T**	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
14	**O**	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
15	**P**	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O

YHHDTXFAAG

(f) a double transposition cipher with first key word BLUE and second key word GOLD.

B	L	U	E
G	O	T	O
B	E	R	L
I	N		

B	E	L	U
G	O	O	T
B	L	E	R
I		N	

GBIOL
OENTR

G	O	L	D
G	B	I	O
L	O	E	N
T	R		

D	G	L	O
O	G	I	B
N	L	E	O
	T		R

ONGLT
IEBOR

2. Using the alpha-numeric substitution below to answer the questions.

original	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
maps to	9	8	7	6	5	4	3	2	1	0	D	C	B	A	H	G	F	E

original	S	T	U	V	W	X	Y	Z	0	1	2	3	4	5	6	7	8	9
maps to	L	K	J	I	P	O	N	M	T	S	Q	R	V	U	Y	X	W	Z

(a) Encrypt the plaintext: RIGHT 10 LEFT 25

E132K STC54 KQU

(b) Decrypt the ciphertext: K5OKT TQXW

TEXTO 0278

3. Decrypt each encrypted text using the specified mechanism.

(a) ciphertext: NJCYQC
encryption mechanism: a Caesar cipher with a shift of 24 (A to Y)

		A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
0	A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
24	Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X

PLEASE

(b) ciphertext: DFTOO ANRBO GCTED
encryption mechanism: a transposition cipher given a row length of 5

15 characters & row length 5 (=5 columns) means 3 rows

D	O	N	O	T
F	O	R	G	E
T	A	B	C	D

DO NOT FORGET ABCD
(we assume ABCD are junk characters but they might not be! So make sure to include them in your decryption)

(c) ciphertext: DMFAT 3RYME IIRS D
encryption mechanism: a transposition cipher given encryption keyword STOP

16 characters and 4 columns means 4 rows.

fill in by columns

O	P	S	T
D	T	M	I
M	3	E	R
F	R	I	S
A	Y	I	D

→

S	T	O	P
M	I	D	T
E	R	M	3
I	S	F	R
I	D	A	Y

read out rows

Midterm 3 is Friday

(d) ciphertext: HYQWQ
encryption mechanism: a shifting substitution cipher / progressive Caesar cipher starting with a shift of 6 (A to G)

BRING

		0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
		A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
6	G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
7	H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
8	I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
9	J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
10	K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
11	L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
12	M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
13	N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
14	O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
15	P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
16	Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P

(e) cipher text: FHHTK ASTL
encryption mechanism: a Vigenère cipher with key word STOP

F H H T K A S T L
S T O P S T O P S
N O T E S H E E T

		0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
		A	**B**	**C**	**D**	**E**	**F**	**G**	**H**	**I**	**J**	**K**	**L**	**M**	**N**	**O**	**P**	**Q**	**R**	**S**	**T**	**U**	**V**	**W**	**X**	**Y**	**Z**
18	**S**	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
19	**T**	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
14	**O**	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
15	**P**	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O

Notesheet

(f) cipher text: OEAOM NTSLD
encryption mechanism: a double transposition cipher with first key word BLUE and second key word GOLD.

Feed in columns →
10 chars, 4 cols ⇒ 2 full rows + 2 characters which go into the * columns
⇒ G, O get 3 chars, D, L get 2
Fill columns

* D	* G	L	* O
O	A	N	S
E	O	T	L
	M		D

→

* G	* O	L	D
A	S	N	O
O	L	T	E
M	D		

→ ASNOO LTEMD

* B	E	* L	U
A	O	L	M
S	O	T	D
N		E	

→

* B	* L	U	E
A	L	M	O
S	T	D	O
N	E		

ALMOST DONE

4. Which of the encryption methods in problems 1 and 2 above preserve letter frequency?

transposition & double transposition preserve letter frequency
shift, sequential shift, substitution, and vigenère do not.

		0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
		A	**B**	**C**	**D**	**E**	**F**	**G**	**H**	**I**	**J**	**K**	**L**	**M**	**N**	**O**	**P**	**Q**	**R**	**S**	**T**	**U**	**V**	**W**	**X**	**Y**	**Z**
0	**A**	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
1	**B**	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
2	**C**	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
3	**D**	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
4	**E**	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
5	**F**	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
6	**G**	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
7	**H**	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
8	**I**	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
9	**J**	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
10	**K**	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
11	**L**	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
12	**M**	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
13	**N**	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
14	**O**	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
15	**P**	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
16	**Q**	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
17	**R**	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
18	**S**	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
19	**T**	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
20	**U**	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
21	**V**	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
22	**W**	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
23	**X**	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
24	**Y**	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
25	**Z**	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y